Bug Identification & Fix Report

Candidate: Ishu Baghel

Assessment: Code Troubleshooting Technical Assessment

1. Node.js Application

Bug 1: Incorrect Stock Adjustment

File: `service/stockservice.js`

Root Cause: Negative adjustments were incorrectly handled using `Math.abs()`.

Fixed Code:

```
let next = book.quantity + delta;
```

Testing: Verified stock increases and decreases correctly.

Bug 2: Category Filtering

File: `repository/bookrepository.js`

Root Cause: The SQL condition used `!=` instead of =.

Fixed Code:

```
if (category) {
  conditions.push("category = ?");
  params.push(category);
}
```

Testing: Verified category-specific results.

Bug 3: Search Missing Latest Character

File: `frontend/js/app.js`

Root Cause: Search used a stale query value.

Fixed Code:

```
searchInput.addEventListener("input", () => {
  const query = searchInput.value.trim();
  pendingQuery = query;
  loadBooks(query);
});
```

Testing: Verified the latest search input was used.

Bug 4: Published-Year Validation Error

File: `Book controller (update function)`

Root Cause: `price` and `publishedYear` arguments were passed in the wrong order.

Fixed Code:

```
const book = bookService.updateBook(
  req.params.id,
  body.title,
  body.author,
  body.isbn,
  body.category,
  body.price,
  body.published_year,
  body.quantity
);
```

Testing: Verified parameter mapping and book updates.

Bug 5: Missing ISBN Column

File: `Books table frontend file`

Root Cause: The ISBN field was missing from the table.

Fix Applied: Added an ISBN column to display each book's ISBN.

Testing: Verified ISBN values appeared in the table.

Bug 6: Search Route Conflict

File: `Express book routes file`

Root Cause: The dynamic `/:id` route was registered before `/search`.

Fixed Code:

```
router.get("/search", controller.search);
router.get("/:id", controller.getById);
router.post("/", controller.create);
router.put("/:id", controller.update);
router.delete("/:id", controller.remove);
router.patch("/:id/stock", controller.adjustStock);
```

Testing: Verified search and book ID endpoints.

2. Python Application

Bug 7: OCR Returning Empty Text

File: `app/recognition/ocr.py`

Root Cause: `--psm 7` assumed a single text line, unsuitable for the tested multi-line image.

Fixed Code:

```
def tesseract_config():
    return "--oem 3 --psm 6"
```

Testing: Verified `POST /api/recognize` returned recognized text.

Bug 8: Empty Text Download

File: `app/utils/files.py`

Root Cause: `w+` truncated the saved text file.

Fixed Code:

```
def read_result(result_id):
    if not is_result_id(result_id):
        raise FileNotFoundError(result_id)

    path = result_dir() / f"{result_id}.txt"

    with path.open("r", encoding="utf-8") as handle:
        return handle.read()
```

Testing: Verified the downloaded text file contained the recognized text.

3. Summary

Application	Bugs Fixed
Node.js	6
Python	2
Total	**8**

All eight fixes were tested. No unresolved issues were identified among the documented bugs. OCR accuracy may vary depending on handwriting and image quality.